

JACOB BAKER-SPITZER

Durham, NC

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Education

Duke University, Pratt School of Engineering

Aug. 2026 – May 2030 (Expected)

Bachelor of Science in Engineering, Electrical & Computer Engineering

Durham, NC

Relevant Coursework

- Matrices and Vectors
- Computing for Engineers
- Engineering Design
- Economic Principles

Experience

Economics Research Project

Summer 2026

Research Assistant

New York, NY

- Built a 1,000+ building dataset spanning about 100 years of NYC construction, later used in an economics research paper, by compiling structural data (floors, elevators, fireproofing) from historical fire records in Google Sheets over 4 weeks.
- Verified lot labels against source records to keep the dataset accurate.

LiRo Group

Aug. 2025

Engineering Shadow

New York, NY

- Documented the roof support systems and structural condition of 3 buildings with photos, notes, and field measurements while shadowing an engineer for one week.

Westchester Youth Soccer League

2023 – 2026

Certified Youth Soccer Referee

Westchester, NY

- Officiated 180+ U9–U12 matches over 3 years, advancing from assistant referee to center referee, by making real-time calls and managing players, coaches, and parents.

Projects

Adaptive Fishing Game (EGR 101) | *Assistive Technology, Design Process*

Fall 2026 – Present

- Defined requirements for a 2-switch fishing game for a child with limited mobility by interviewing the client and his care team with a 5-person design team.

Hands-On Engineering Builds (EGR 101) | *Soldering, 3D Printing*

Fall 2026

- Soldered an LED switching circuit for a lighted sign and 3D-printed a custom mug with the Duke logo.

Technical Skills

Languages: Python

Developer Tools: VS Code, Git/GitHub, Arduino IDE, Onshape (CAD), Google Sheets

Electronics & Prototyping: Soldering, Circuit Prototyping, 3D Printing

Leadership / Extracurricular

Riverdale Kenshikai Karate

Aug. 2013 – Present

Black Belt (First Degree) & Assistant Instructor

Bronx, NY

- Earned a black belt in over 13 years of training and 600+ classes.
- Co-taught 50+ classes of 10–20 student alongside other black belts by leading drills, correcting form, and preparing students for belt promotions.

Catalyst

Sep. 2026 – Present

New Member Education Cohort

Durham, NC

- Selected as 1 of 20 from roughly 200 applicants through a multi-round recruitment process.

Ignite

Spring 2027

Incoming Engineering Trainer, Light/Energy Team

Durham, NC